

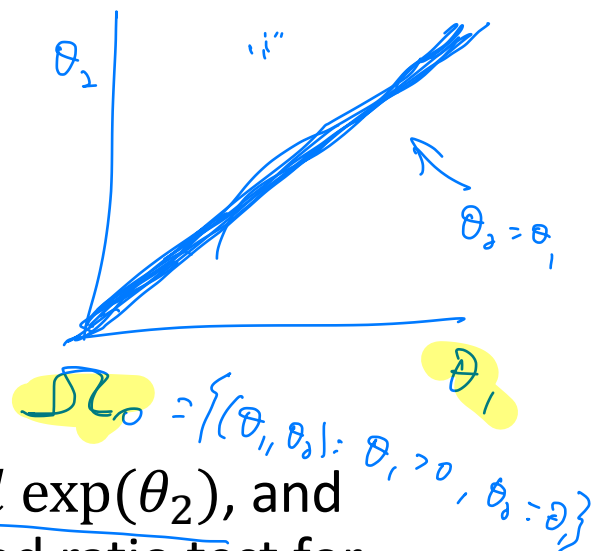
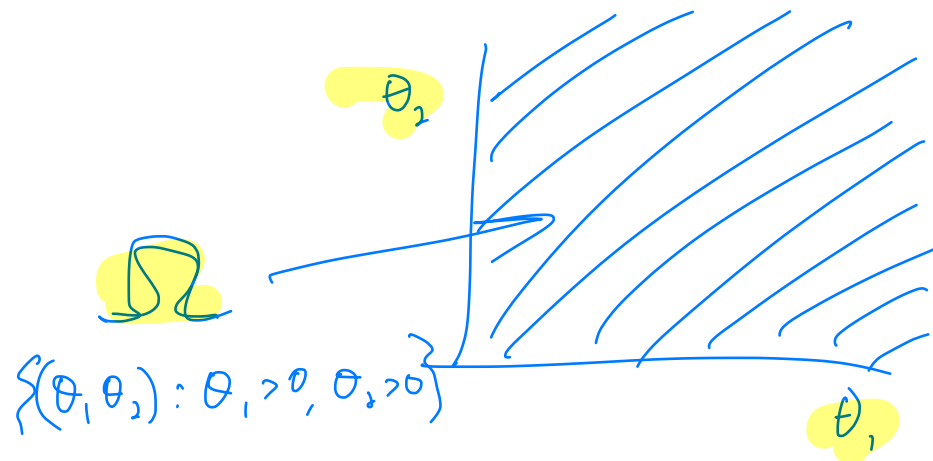
# HYPOTHESIS TESTING

Likelihood ratio tests using approximate distribution

# Likelihood ratio tests – approximate distribution

- Many of the “standard” tests (t-tests,  $\chi^2$  tests for variance,  $F$  test for comparing two variances) are equivalent to their corresponding likelihood ratio tests.
- In such situations, the rejection criterion  $\lambda < k$  exactly corresponds to a decision based on a test statistic with a known distribution.
- But in many other cases, the likelihood ratio test cannot be expressed in terms of some parameter with a “known” distribution.
- Theorem: For large  $n$ , the test statistic  $-2 \log \lambda$  has an approximate  $\chi^2$  distribution with  $q$  degrees of freedom, where  $q$  is the number of restrictions imposed on the parameter space  $\Omega_0$

2-dimensional



Example:  $X_1, X_2, \dots, X_m \sim iid \exp(\theta_1)$ ,  $Y_1, Y_2, \dots, Y_n \sim iid \exp(\theta_2)$ , and the two samples are independent. Derive the likelihood ratio test for  $H_0: \theta_1 = \theta_2$  vs  $H_a: \theta_1 \neq \theta_2$ .

$$f_X(x) = \frac{1}{\theta_1} e^{-\frac{x}{\theta_1}} \quad f_Y(y) = \frac{1}{\theta_2} e^{-\frac{y}{\theta_2}}$$

$$L(\theta_1, \theta_2) = \prod_{i=1}^m \frac{1}{\theta_1} e^{-\frac{x_i}{\theta_1}} \prod_{j=1}^n \frac{1}{\theta_2} e^{-\frac{y_j}{\theta_2}} = \theta_1^{-m} e^{-\frac{1}{\theta_1} \sum x_i} \theta_2^{-n} e^{-\frac{1}{\theta_2} \sum y_j}$$

$$= \theta_1^{-m} \theta_2^{-n} e^{-\frac{m\bar{x}}{\theta_1}} e^{-\frac{n\bar{y}}{\theta_2}}$$

maximize over  $\Omega$  (unrestricted)

$$l(\theta_1, \theta_2) = \log L(\theta_1, \theta_2) = -m \log \theta_1 - \frac{n \log \theta_2}{\theta_1} - \frac{m\bar{x}}{\theta_1} - \frac{n\bar{y}}{\theta_2} \quad \text{MLE}$$

$$\frac{\partial}{\partial \theta_1} l = -\frac{m}{\theta_1} + \frac{1}{\theta_1^2} m\bar{x} \stackrel{!}{=} 0 \quad -m + \frac{m\bar{x}}{\theta_1} = 0 \quad \frac{\bar{x}}{\theta_1} = 1 \quad \boxed{\hat{\theta}_1 = \bar{x}} \quad \text{check 2nd deriv...}$$

Similarly,  $\hat{\theta}_2 = \bar{Y} \leftarrow$  unrestricted MLEs

maximize  $L$  over  $\Omega_0$   $\theta_1 = \theta_2$

$$\frac{\sum x_i + \sum y_i}{m+n}$$

$$l = \log L(\theta_1, \theta_2) = -m \log \theta_1 - \frac{m\bar{X}}{\theta_1} - n \log \theta_1 - \frac{n\bar{Y}}{\theta_1}$$

$$\frac{\partial}{\partial \theta_1} l(\theta_1, \theta_1) = -\frac{m}{\theta_1} + \frac{m\bar{X}}{\theta_1^2} - \frac{n}{\theta_1} + \frac{n\bar{Y}}{\theta_1^2} \stackrel{\text{set}}{=} 0$$

$$\hat{\theta}_1 = \frac{m\bar{X} + n\bar{Y}}{m+n} = \hat{\theta}_2$$

restricted

Solve for  $\theta_1$

$$L(\hat{\Omega}_0) = \hat{\theta}_1^{-(m+n)} e^{-\frac{1}{\hat{\theta}_1}(m\bar{X} + n\bar{Y})} = \hat{\theta}_1^{-(m+n)} e^{-\frac{(m\bar{X} + n\bar{Y})}{m+n} \cdot \frac{m+n}{m\bar{X} + n\bar{Y}}}$$

(restricted)

$$= \hat{\theta}_1^{-(m+n)} e^{-(m+n)}$$

$$L(\hat{\Omega}) = \bar{X}^{-m} \bar{Y}^{-n} e^{-(m+n)}$$

$$LR: \lambda = \frac{L(\hat{\Omega}_0)}{L(\hat{\Omega})} = \frac{\left(\frac{m\bar{X} + n\bar{Y}}{m+n}\right)^{-(m+n)} e^{-(m+n)}}{\bar{X}^{-m} \bar{Y}^{-n} e^{-(m+n)}}$$

Reject  $H_0$  if  $\lambda$  is "small"

dist ???

Reject  $H_0$  if  $\underbrace{-2 \log \lambda}_{\sim \chi^2_1} > \chi^2_{1, 0.05}$

Approx. test  
need  $m$  &  $n$  to  
be "large"